

PETERHEAD HARBOUR, United Kingdom

WMO: 030920

Lat: 57.502N

Lon: 1.774W

Elev: 49

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.5	33.3	24.7	18.8	36.2	26.5	20.5	36.5	33.5	42.4	31.7	41.4	9.9	270	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	64.6	58.9	62.8	57.6	61.3	57.0	60.3	62.7	59.2	61.3	58.2	60.2	8.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	75.2	61.5	58.2	72.5	60.4	57.2	69.9	59.4	26.6	63.1	25.9	61.4	25.3	60.4	65.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.1	26.9	24.1	DB	28.2	69.6	3.1	2.9	25.9	71.7	24.1	73.4	22.3	75.0	20.1	77.1	(4)
(5)			WB	26.9	63.4	2.8	1.5	24.8	64.5	23.2	65.4	21.6	66.3	19.6	67.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	41.8	41.4	42.7	45.5	49.2	53.4	56.8	57.5	55.1	50.7	46.4	42.6	(6)	
	DBStd	6.77	3.38	4.14	4.06	3.50	3.43	2.91	2.57	2.61	3.04	3.95	3.76	4.60	(7)	
	HDD50	1314	254	243	227	141	55	4	0	0	3	41	115	231	(8)	
	HDD65	5975	719	661	690	585	491	348	255	231	298	445	557	695	(9)	
	CDD50	815	0	2	2	6	30	106	210	234	155	61	8	1	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	11.2	14.1	13.8	12.8	10.9	9.1	9.4	7.7	7.9	10.2	12.4	12.9	13.3	(14)	
(15) Precipitation	PrecAvg	30.7	2.5	2.4	2.2	2.0	2.1	2.3	2.4	2.2	2.6	3.8	3.5	2.9	(15)	
	PrecMax	39.1	4.8	4.4	4.8	5.2	4.6	4.1	5.1	4.1	8.5	7.3	7.0	4.9	(16)	
	PrecMin	22.2	1.1	0.5	0.7	0.9	0.5	0.5	0.8	0.6	0.6	0.6	2.2	1.0	(17)	
	PrecStd	4.3	1.0	1.1	0.9	1.1	1.1	1.1	1.5	0.9	1.7	1.7	1.4	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	56.2	55.9	59.0	61.4	64.4	67.7	67.4	64.7	61.2	57.1	53.6	(19)	
		MCWB	49.1	51.5	50.3	50.8	54.4	58.0	61.7	60.8	58.5	55.7	54.1	50.3	(20)	
	2%	DB	49.8	52.3	52.3	54.4	57.7	61.1	64.1	64.7	62.5	58.1	54.5	51.3	(21)	
		MCWB	47.0	48.9	47.4	48.8	51.9	55.2	58.9	59.1	57.3	54.2	51.1	48.7	(22)	
	5%	DB	48.1	49.3	50.1	52.3	55.6	59.3	62.0	62.9	60.8	56.6	53.0	49.9	(23)	
		MCWB	45.6	46.3	46.1	47.7	50.5	53.7	57.4	57.9	56.4	53.6	50.5	47.7	(24)	
	10%	DB	46.7	47.1	48.2	50.5	53.8	57.6	60.5	61.4	59.2	55.6	51.7	48.7	(25)	
		MCWB	44.7	44.1	44.8	46.6	49.6	53.0	56.9	57.5	55.9	52.9	49.3	46.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	52.1	51.1	52.8	55.1	58.9	62.6	62.3	60.2	56.9	54.3	50.9	(27)	
		MCDB	52.2	55.3	54.6	56.9	60.0	62.8	66.3	65.7	62.8	59.1	55.6	52.5	(28)	
	2%	WB	47.7	49.3	48.6	50.0	53.0	56.7	60.0	60.6	58.5	55.5	52.1	49.3	(29)	
		MCDB	49.3	51.9	51.6	53.0	56.6	59.7	62.4	63.0	60.8	57.1	53.6	50.8	(30)	
	5%	WB	46.2	46.7	46.7	48.5	51.8	55.3	58.8	59.6	57.5	54.5	51.0	48.0	(31)	
		MCDB	47.5	49.0	49.0	51.0	54.3	57.8	60.8	61.5	59.4	56.0	52.6	49.5	(32)	
	10%	WB	44.9	44.6	45.3	47.3	50.6	54.1	57.7	58.6	56.7	53.6	49.7	46.8	(33)	
		MCDB	46.7	46.5	47.4	49.6	52.9	56.3	59.6	60.3	58.5	55.1	51.3	48.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.3	6.6	6.6	6.7	6.6	6.0	6.6	6.1	6.0	5.9	5.6	(35)	
		MCDBR	7.3	9.0	10.0	10.0	9.6	9.3	8.7	8.9	8.3	7.2	7.1	6.4	(36)	
	5% WB	MCWBR	7.0	7.8	7.0	6.4	5.9	5.6	5.5	5.8	5.9	6.1	7.0	6.3	(37)	
		MCWBR	6.5	8.6	9.0	8.3	7.9	7.7	7.4	7.5	6.2	5.9	6.1	6.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.315	0.339	0.362	0.386	0.377	0.378	0.383	0.372	0.358	0.343	0.319	0.298	(40)	
		taud	2.240	2.324	2.329	2.292	2.378	2.405	2.398	2.437	2.474	2.416	2.282	2.177	(41)	
	Ebn at Noon		169	221	249	261	272	273	269	264	251	218	167	136	(42)	
		Edh at Noon	16	23	30	36	35	34	34	30	25	21	15	13	(43)	
(44) All-Sky Solar Radiation	RadAvg		131	328	698	1174	1605	1580	1535	1229	824	421	170	87	(44)	
	RadStd		16	32	67	106	86	124	107	84	66	35	15	10	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page